

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 5

7 Introduction to Computer Architecture (swm11)

A microprocessor design team currently has a classical five-stage pipelined processor comprising: instruction fetch, decode + register fetch, execute, data memory access, write-back. They also included independent instruction and data level-1 caches (32 KiB each) and one level-2 cache (256 KiB).

Pipelining

- (a) In the context of this five-stage pipeline, explain what the following hazards are and how they might arise:

- (i) control hazards, [2 marks]

Answer: Control hazards arise when the control-flow of the program changes as a result of a branch, jump, exception or interrupt. For this five-stage pipeline, instruction fetch will sequentially fetch the next instruction until instructed to do otherwise. If branch and jump instructions are executed in the execute stage, then two instructions after the branch/jump will already have been fetched in error and must not be executed if the branch/jump is taken.

- (ii) structural hazards. [2 marks]

Answer: Structural hazards arise when functional units are shared. For example, if instruction-fetch and memory-access stages shared the same single ported memory then this would be a structural hazard since they would need to take turns to use the shared resource. However, typically instruction-fetch and memory-access have their own independent caches, which eliminates the structural hazard. There are no structural hazards in this simple pipeline.

Pipelining

- (b) For this five-stage pipeline, what are forwarding paths and why they are needed? Include a sketch of the pipeline indicating where the forwarding paths are located. [3 marks]

Answer: Forwarding paths are needed to help mitigate data hazards, which arise when stale register values are read from the register file. Instead of blocking a dependent instruction in the decode stage until the register it requires is written back, forwarding paths allow recently produced values to be made available earlier.

For the sketch I expect to see the results from execute and memory-access stages being made available as inputs to execute.

Pipelining;
caching

- (c) To boost performance, new 128 KiB level-1 instruction and data caches are proposed, but these need to be pipelined to maintain the clock frequency resulting in 2-cycle access latency. The level-2 cache is boosted to 1 MiB without increasing the level of pipelining. This takes the five-stage pipeline to seven-stages.

- (i) Sketch the new pipeline. Why might new forwarding paths be required? [3 marks]

Answer: The new 7-stage pipeline:

IF0	IF1	DC	EX	MA0	MA1	WB
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Where:

- IF0 and IF1 are two stages of instruction fetch
- DC = decode and register fetch
- EX = execute
- MA0 and MA1 are the two stages of memory access
- WB = write-back

A new forwarding path will be required from MA0 back to execute. While this cannot forward an early result from memory access, it is required to forward a recent result produced by the EX stage.

- (ii) What is a load-to-use penalty for the five-stage and seven-stage pipelines? [2 marks]

Answer: For the original pipeline with appropriate forwarding logic, the load-to-use penalty is one clock cycle. This will increase to two clock cycles for the seven-stage pipeline.

- (iii) What is a branch penalty for the five-stage and seven-stage pipelines? [2 marks]

Answer: If branches are taken in the execute stage then the original pipeline has a two clock cycle penalty. For the seven-stage pipeline this increases to three clock cycles.

- (iv) What is the cache access latency in cycles for each design given the following table of parameters? [4 marks]

	pipeline size	
	5-stage	7-stage
Level-1 data-cache miss rate	5%	3%
Level-2 cache miss rate	2%	1.5%
Level-1 data-cache access latency in cycles	1	2
Level-2 cache access latency in cycles	9	9
DRAM access latency in cycles	100	100

Answer:

let $mr1$ = miss-rate for the level-1 data cache
 $mr2$ = miss-rate for the level-2 data cache
 $al1$ = access latency for the level-1 data cache
 $al2$ = access latency for the level-2 cache
 dl = DRAM access latency

— *Solution notes* —

$$\text{Access latency} = al1 + mr1 \times (al2 + mr2 \times dl)$$

So for the 5-stage pipeline the access latency = $1 + 5\% \times (9 + 2\% \times 100) = 1.55$ cycles

So for the 7-stage pipeline the access latency = $2 + 3\% \times (9 + 1.5\% \times 100) = 2.315$ cycles

Executable
processor models

- (d) What is tandem verification and how could the technique be used to test the pipelined processor designs? [2 marks]

Answer: Tandem verification is testing technique where the stream of completed instructions from a processor is compared against a known good model or other processor implementation. The test coverage is only as good as the test program being executed.
